

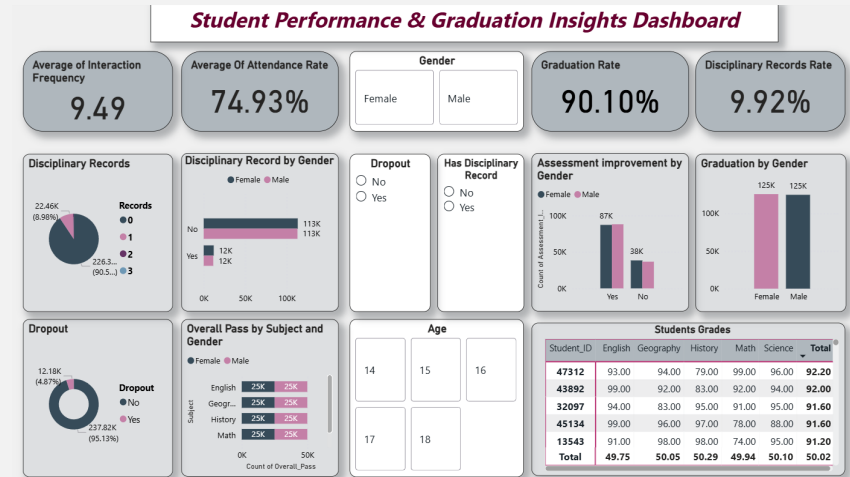
Student Performance Analysis & Graduation Insights

Leveraging MySQL and Power BI for Data-Driven Education

Supervised by: Eng. Waleed Mohammed

Presented by: Data Pioneers Team

Members: Shady Abdallah, Sohaila Ahmed



Project Introduction & Significance

Education has always been a key driver of human and societal development. In today's data-driven era, analyzing student performance has become **crucial for understanding learning behavior**, identifying areas of improvement, and optimizing educational systems globally. This project leverages data analytics to transform academic grades and related indicators into actionable insights.

Importance of Performance Analysis

- | | | | |
|---|---|---|--|
| <ul style="list-style-type: none">• Detect Learning Gaps: | Identify challenges faced by students and provide timely, targeted interventions. | <ul style="list-style-type: none">• Support Decision-Making: | Provide evidence-based insights for curriculum design and resource allocation. |
| <ul style="list-style-type: none">• Enhance Personalized Learning: | Tailor instruction and support to meet individual student needs effectively. | <ul style="list-style-type: none">• Improve Institutional Performance: | Foster accountability and improve overall academic outcomes. |

Data Preparation: Key SQL Operations

The project utilized SQL for robust data cleaning and the creation of analytical views to ensure data quality and aggregate key performance indicators.

1. Data Standardization (Example: Gender Field)

```
UPDATE student_performance
SET Gender = CASE
WHEN LOWER(Gender) IN ('male', 'm') THEN 'Male'
WHEN LOWER(Gender) IN ('female', 'f') THEN 'Female'
ELSE 'Unknown'
END;
```

2. Unified KPI Dashboard View Creation

```
CREATE OR REPLACE VIEW student_kpi_dashboard AS
SELECT
ROUND(AVG((Math + Science + English + History + Geography) / 5), 2) AS Avg_Grade,
ROUND(AVG(Attendance_Percentage), 2) AS Avg_Attendance,
ROUND(SUM(CASE WHEN Graduation = 'Yes' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS Graduation_Rate
FROM student_performance;
```

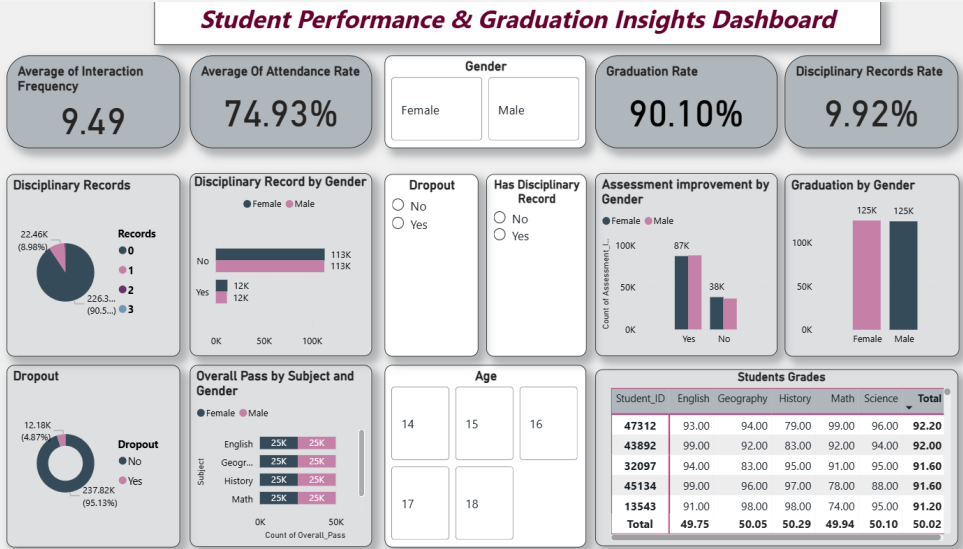
The Main Dashboard: Comprehensive Performance View

Student Performance & Graduation Insights Dashboard

The developed dashboard is an interactive visual report that summarizes student progress and learning patterns using multiple Key Performance Indicators (KPIs) and analytical charts. It enables educators and administrators to explore performance by gender, attendance, study hours, subjects, and graduation status.

Key Metrics at a Glance:

- Average Attendance Rate: 74.93%
- Graduation Rate: 90.10%
- Average Interaction Frequency: 9.49
- Dropout Rate: 4.87%



Deep Dive: Drill-Through Analysis

(Student 1)

The dashboard includes a **Drill-Through** feature to examine individual student records and provide targeted insights for educational interventions.

Demographics

Age: 17 years old

Gender: Female

Engagement Metrics

Interaction Frequency:16 (High engagement)

Academic Status

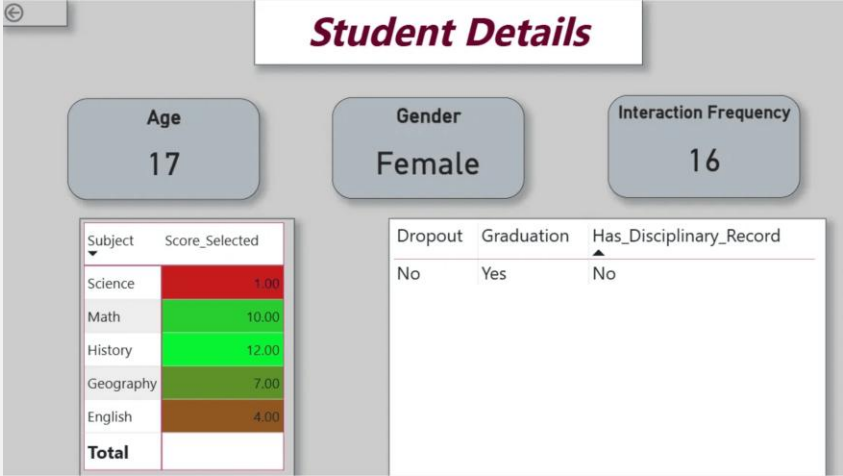
Dropout Status: No

Graduation: Yes

Disciplinary Record: No

Key Insight

This student demonstrates **high engagement** but shows **critically low academic scores** across all subjects. This discrepancy suggests a need for targeted academic support and intervention strategies.



Comparative Student Insights and Conclusion

Student 2 Details (Drill-Through View)

Demographics: Age 17, Male **Engagement:** Low Interaction Frequency (5)

Status: No Dropout, Graduated (Yes) **Discipline:** No Disciplinary Record

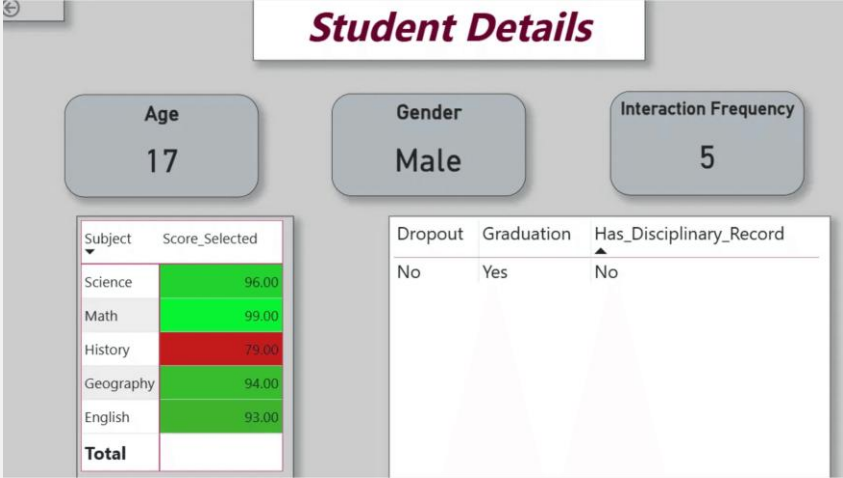
Performance: This student demonstrates **high scores in most subjects** (Science 96.00, Math 99.00), with a notable dip in History (79.00). Despite lower interaction frequency compared to Student 1, this student achieved strong academic results.

Key Insight: Contrasting Patterns

The comparison between Student 1 (high engagement, low scores) and Student 2 (low engagement, high scores) highlights the **complexity of student performance**. This demonstrates the critical value of the drill-through feature for identifying individualized intervention strategies.

Project Conclusion

The **Student Performance Analysis** project successfully integrates data preparation (MySQL) and visualization (Power BI) to provide a powerful tool for educators. The main dashboard offers a holistic view of institutional performance, while the drill-through feature enables **personalized, data-driven interventions** for every student.



Conclusion & Proposed Solutions for Student Performance

Proposed Solutions to Raise Performance Rates

****Performance Challenge:****

Analysis showed a discrepancy between engagement indicators and academic grades.

Some students show high engagement rates (e.g., Student 1) but suffer from low grades, suggesting the problem lies in the effectiveness of study methods or the need for specialized academic support.

****Importance of Individual Analysis:****

The "Drill-Through" feature confirmed that general indicators can be misleading, and effective intervention requires a precise understanding of the root causes of problems at the individual student level.

Proposed Solution	Implementation Action	Goal of the Action
Targeted Interventions	Use dashboard data to identify students at risk of failure or academic decline and provide them with customized study plans.	Ensure every student receives the appropriate support for their individual needs.
Academic Mentorship Programs	Connect low-performing students with specialized mentors or teachers to provide individual or small-group tutoring sessions.	Provide necessary guidance to improve study skills and time management.
Improve Teaching Methods	Analyze data to identify subjects with the lowest average grades and develop more engaging and interactive teaching methods for these subjects.	Raise the level of understanding and comprehension in difficult subjects.

Project Team & Credits



Shady Abdallah

Data Analysis & Visualization

Specialized in data preparation, SQL optimization, and Power BI dashboard design for educational analytics.



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Sohaila Ahmed

Project Lead & Research

Led market research, data analysis strategy, and comprehensive documentation for the Student Performance Analysis project.



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Student Performance Analysis & Graduation Insights

A comprehensive data-driven solution for educational institutions

Supervised by Eng. Waleed Mohammed